

EXP2-4 Blamestorming Part 2

To determine the factors affecting the strength of The Quakeproof Bearings Company's rubber material, the following experiments were conducted.

Natural rubber, sulphur and accelerator were weighed out in appropriate proportions and compounded in a 2-roll mill. Samples of each formulation were moulded into flat sheets in a compression moulder and tested in tension. The elastic modulus at 100% strain was recorded for each sample. Each measurement was repeated 3 times to assess the random variation in the measurement for each formulation.

The factors tested were the mould temperature, the type of accelerator used in the rubber, the sulphur content and the moulding time. Each factor was set at two levels: Mould temperatures were 155°C or 170°C. Accelerators from both suppliers were used, A1 was the old accelerator and A2 was new accelerator. The sulphur crosslinker was added at 2 parts per hundred and 4 parts per hundred. Mould dwell times were 20 minutes or 40 minutes. These are summarized in Table 1.

Table 1. Factors and Level used in the factorial designed experiment

Factors		Levels	
		+	-
A	Temperature	155	170
B	Accelerator	A1	A2
C	Sulphur	2 pph	4 pph
D	Mould Time	20 min	40 min

16 experiments were performed according to a factorial experimental design as detailed in the table below. Three measurements of tensile 100% strain modulus were made for each run. The results are shown in the table below.

Table 2. Factorial design and output measurements for each 'Run' of the design.

Factors					Outcomes (Tensile modulus @ 100% strain, MPa)		
					1	2	3
Run 1	+	+	+	+	1.15	1.22	1.12
Run 2	-	+	+	+	0.73	0.74	0.69
Run 3	+	-	+	+	1.33	1.25	1.22
Run 4	-	-	+	+	1.16	1.06	1.03
Run 5	+	+	-	+	0.75	0.71	0.72
Run 6	-	+	-	+	1.35	1.50	1.47
Run 7	+	-	-	+	0.66	0.69	0.74
Run 8	-	-	-	+	1.50	1.46	1.48
Run 9	+	+	+	-	1.13	1.22	1.17
Run 10	-	+	+	-	0.71	0.69	0.74
Run 11	+	-	+	-	1.31	1.26	1.23
Run 12	-	-	+	-	1.15	1.09	1.06
Run 13	+	+	-	-	0.70	0.73	0.74
Run 14	-	+	-	-	1.51	1.33	1.46
Run 15	+	-	-	-	0.70	0.74	0.66
Run 16	-	-	-	-	1.51	1.49	1.46

It took 1 person 1 day to make all the samples and 1 day to test them. i.e. 2 man/days.

The resources required were: 1 balance, 1 compounding mill, 1 compression moulder, 1 tensile testing machine, 1 person.

What you should do:

1. Using the data given above, **every group member** should analyse the test results. You can copy and paste the data directly from this Word document on the QXU5017 QMPlus pages into Excel to do the calculations (see the notes given below). By running the analysis from the averaged strength values, the parameters that affect the strength can be found. By doing the same analysis for the range values, the parameters that affect the reliability can be found.
2. Hold a **group management meeting** to discuss the results and make an action plan to complete the work. Look carefully at the results of the analysis – what do they tell you? Then, **the whole group** together should compare the results you obtained (did everyone get the same answers? - check for mistakes) and discuss which factors, or combination of factors, have actually caused the new material to change its properties. What causes the product to perform badly and how might that come about? Look again at the Blamestorming Conversation in the meeting at Quakeproof Bearing Co. How do the results of the factorial experiment help us to interpret the evidence presented in that conversation? Can you now see where there was potential for the process to go wrong?
3. The group should then **write a brief report** (max. 3 pages total) covering the following 5 points:
 - 1) Present the results of the calculations;
 - 2) State which factors can actually cause the process to change.
Which factors or interactions are important and which are not important?
 - 3) Discuss how the results support what was said in the *Blamestorming Conversation*;
 - 4) Consider if your Group's experimental plan would have found this out – be critical;
 - 5) Suggest what should be done next to fix the problem.

You do not need to describe the background and methods again – reference this document from the QXU5017 QMPlus page.

The **title** of the report will be: **EXP2-4 Part 2 Report XX** (where XX is your group number)

The names of **all the group members** should be given as authors of the report.

You will **submit** your group's report as a **pdf document on QMPlus** using the file name:

'QXU5017 EXP2-4 Part 2 Report XX' (where XX is your group number).

The report must be completed and **submitted by 24:00 on Friday 23rd May 2025**.

The report must be the **work of the whole group** and the content agreed by all group members.

All group members must hit the submit button to submit the report.

4. Group Peer Assessment:

Your final score for this exercise will be calculated from the **Group Mark** multiplied by the **Group Peer Assessment Mark**. The **Group Peer Assessment Mark** is determined by **you**.

All **Group Peer Assessment** forms must be completed **by 24:00 31st May** or they won't be included.

Schedule for EXP2-4 Part 2**M Groups**

Date:	Time:	Place:	Activity:
Friday 16th	08:30-10:10	A310	Introduction to Part 2 of EXP2-4
Monday 19th	10:30-12:10	A310	Data processing & analysis
Wednesday 21st	10:30-12:10		Blamestorming
Friday 23rd	08:30-10:10		Complete and Submit Report

P Groups

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Notes:

- The factorial design experiment is not applicable to all experimental situations but can be most useful where many variables affect a process and there may be interaction between them. E.g., such methods are very useful in understanding process control variables, and are used widely in industry. There are many other types of experimental design that may be more appropriate in other situations (e.g., Taguchi designs for design processes).
- The factorial design allows all factors to be investigated simultaneously without trying to guess which is important or weighting one factor more than another.
- For several factors and several repeats of each run, the differences between random variations and genuine effects can be identified. This can allow assessment of the factors affecting the process and those factors affecting reliability.

Procedure:

For each **Run**,

calculate the **average outcome** for each test run, and note the **range** of values (max-min)

For each **Factor**, SUM all the averaged values when the factor is + **(SUM +)**

SUM all the averaged values when the factor is – **(SUM -)**

Find the **Difference** between SUM+ and SUM-

The “**Main Effect**” is the Difference divided by the number of values in each SUM (8 in this case)

Interactions can be plotted, or calculated by adding the signs together for each pair of factors

e.g. + and + = +,

+ and - = -,

- and - = + forming a new pattern of + and -

This pattern can now be evaluated in the same way as above.

Interactions between more than 2 factors are very rare – **do not consider them here.**

Effects must be greater than **Random Variations** in the test methods. Random variation can be found from the average **Range** of the data, multiplied by 2 and divided by the number of values in each SUM (8 in this case). If the **Effect** or the **Interaction** is much greater than the **Random Variation** then it has a real effect on the process.